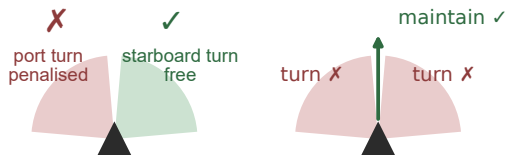


(a)

Directional non-compliance penalty

Head-On · Give-Way · Overtaking

Stand-On

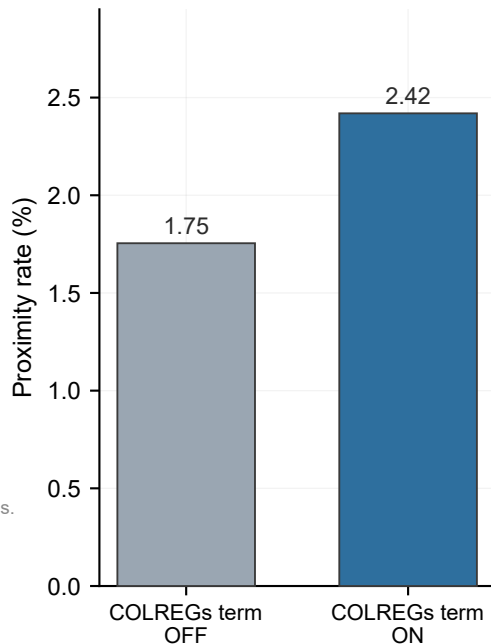


$$R_i^{\text{COLREGs}} = -\alpha_c (1 + \rho_i) \mathbf{1}[\rho_i > 0.3] (\dots)$$

A directional penalty on non-compliant turning — not a positive compliance bonus.
Weighted by collision risk ρ_i and applied only above a risk gate.

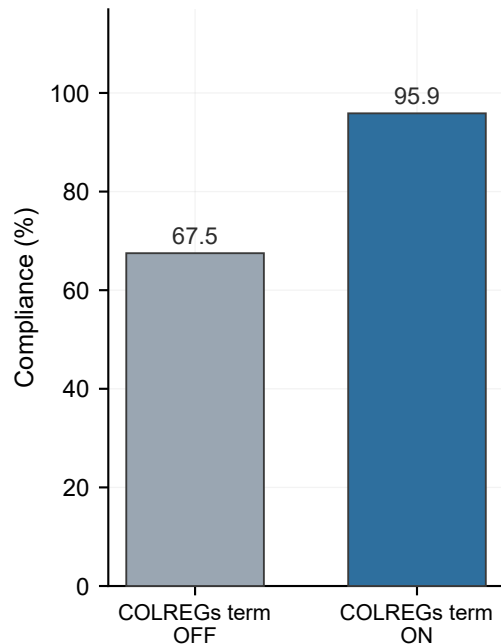
(b)

Proximity rate



(c)

Overall COLREGs (PRIMARY v2)



3 seeds. The COLREGs term raises PRIMARY v2 compliance (sit2/sit3 especially) and is the only ablation that moves C off the ceiling. Proximity is not improved by the term (OFF can look safer while violating stand-on/give-way). Pick training by C here, not by proximity.